ANTEATER CHESS
USER MANUAL
VERSION 1.0

TEAM 21
Naveen Khan
Mia Ness
Austin Nguyen
Queency Tan
Aman Macwan
Gurvin Dhanjal

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GLOSSARY

ANT - ANOTHER NAME FOR A PAWN IN ANTEATER CHESS

ANTEATER - PIECE UNIQUE TO ANTEATER CHESS. MOVES ONE SQUARE IN ANY DIRECTION, LIKE A KING. THE ANTEATER IS ONLY ALLOWED TO CAPTURE PAWNS, WITH THE ABILITY TO CATCH MULTIPLE IN ONE MOVE.

BISHOP - PIECE THAT MOVES DIAGONALLY ACROSS THE BOARD ANY NUMBER OF SQUARES.

CHECK - A STATE IN WHICH A PLAYER'S KING IS UNDER DIRECT ATTACK BY AN OPPOSING PIECE. THE PLAYER MUST RESOLVE THE CHECK ON THEIR NEXT MOVE.

CHECKMATE - A CHECK FROM WHICH A KING CANNOT ESCAPE.

KING - MOVES ONE SQUARE IN ANY DIRECTION. THE GAME ENDS WHEN A KING IS CHECKMATED.

KNIGHT - PIECE THAT MOVES IN AN L-SHAPE: TWO SQUARES IN ONE DIRECTION AND ONE SQUARE PERPENDICULAR. ONLY PIECE WITH THE ABILITY TO JUMP OVER OTHER PIECES.

PAWN - ANOTHER NAME FOR AN ANT IN ANTEATER CHESS. MOVES FORWARD ONE SQUARE (TWO IF MOVING FROM STARTING POSITION) AND CAPTURES DIAGONALLY. IF A PAWN REACHES THE OPPOSITE END OF THE BOARD, IT IS PROMOTED TO ANOTHER PIECE OF CHOICE.

QUEEN - PIECE THAT CAN MOVE HORIZONTALLY, VERTICALLY, AND DIAGONALLY ANY NUMBER OF SQUARES.

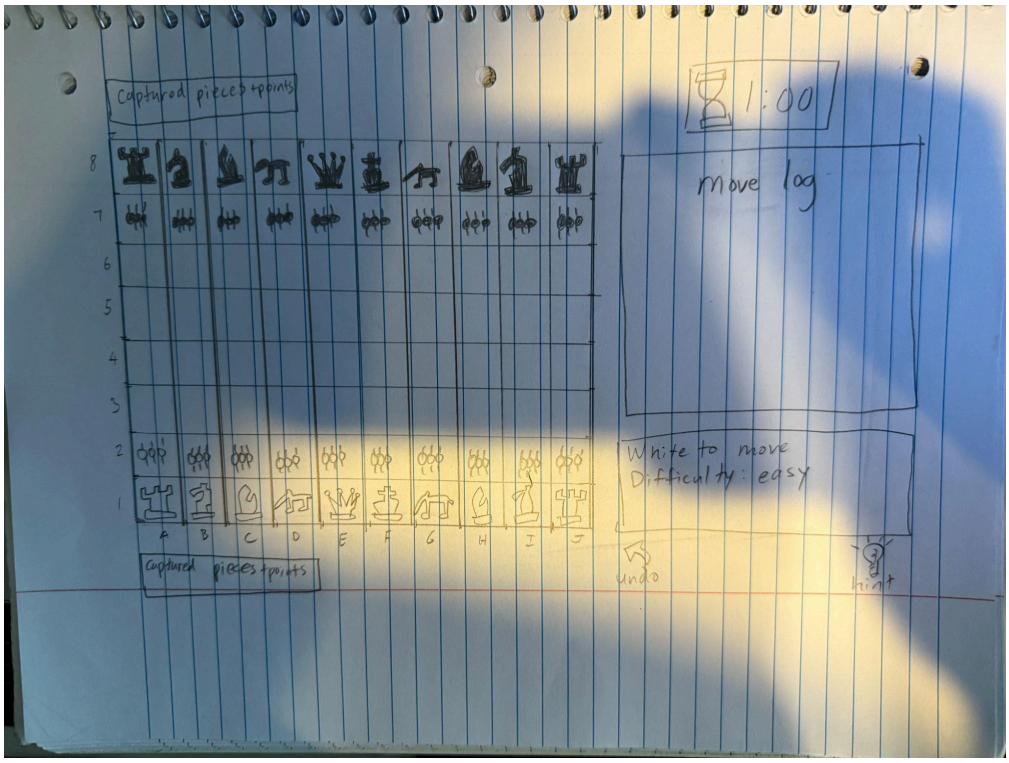
ROOK - PIECE THAT MOVES HORIZONTALLY AND VERTICALLY ANY NUMBER OF SQUARES.

CASTLE - WHEN THERE ARE NO PIECES IN BETWEEN THE KING AND ROOK, THE KING MOVES TWO SPACES TOWARDS THE ROOK, AND THE ROOK MOVES ONE SQUARE TO THE OTHER SIDE OF THE KING

PROMOTION - A PAWN THAT REACHES THE BACK ROW OF THE BOARD CAN BE PROMOTED TO ANY PIECE THAT IS NOT A PAWN

1.1 Usage Scenario

The user launches the Anteater Chess executable. An initial screen asks the user to choose which color (white or black) they would like to use and which difficulty level they would like to play on (easy, medium, or hard). The user selects white and easy and clicks “start game.” The main screen shows the 8x10 chess board with the user to move first since they chose white. A timer counts down from 1 minute. The user clicks on an ant at F2 and the two squares in front of it are highlighted to indicate the possible moves. The user moves the ant to F4 and the move is added to the move log. The computer then moves an ant from F7 to F5. The user clicks the “hint” button to display a possible legal move. Later in the game, the user uses their anteater piece to eat a row of the computer’s ants that are sitting next to each other. Eventually, the user puts the computer in checkmate and the program displays a message that reads “White wins!” with the option to start a new game or quit.



1.2 Goals

The goal of our chess program is to simulate a chess game with a programmed opponent. The program will automate one side of the game so a human can play against it. The automated side should only be able to play legal moves, which should have some level of strategy behind them. The game should limit each player's turn to one minute and signal when a player is in check and will win. The program should highlight all possible moves when a certain chess piece

is clicked and should not allow the human user or the computer opponent to make moves outside of the capabilities of each piece.

1.3 Features

Anteater Chess provides a rich set of features which deliver an engaging and complete chess experience with a twist. The following features are included in Version 1.0:

Custom Anteater Piece (Reference Image 1.31)

- Unique piece that moves one square in any direction (like a King), but is restricted to capturing Ants (Pawns) only.
- The Anteater piece can capture multiple Ants in a single move if they're orthogonally adjacent to one another, making it a powerful offensive tool.
- Ants that are captured by the Anteater are removed from the board simultaneously with the move.

8x10 Game Board (Reference Image 1.32)

- The game is played on an expanded 8x10 board (versus the typical 8x8) to accommodate the additional Anteater pieces on each side.
- Board columns are labeled A-J and rows are numbered from 1-10 for standard algebraic notation.

AI Computer Opponent with Difficulty Levels

- The program includes a computer controlled opponent that plays legal strategic informed moves automatically.
- There are three difficulty levels available: Easy, Medium, Hard, allowing players of all skill levels to enjoy the game.
- The player may choose to play as White or Black at the beginning of the game.

Move Highlighting and Legal Move Enforcement (Reference 1.33)

- When a piece is selected, all legal destination squares are highlighted on the board, making it easy to plan moves.
- The program strictly enforces chess rules, neither the human player nor the computer can make an illegal move.
- A hint button is available so the player can request a suggested legal move at any point during their turn.

Per-Turn Timer

- Each player's turn is limited to one minute. A visible countdown timer tracks the remaining time for the active player.
- The timer automatically stops when a move is made and resets for the opposing player's turn.

Move Log and Captured Piece Tracking

- A running move log records every action taken during the game for both players.

- Captured pieces are displayed on-screen and recorded in a text file for reference.
- A point counter tracks each player's total score based on the value of captured pieces.

Check, Checkmate, and Stalemate Detection

- The program automatically detects when a player's King is in check and signals this to both players.
- Checkmate is recognized and triggers the end-of-game message, identifying the winner.
- Stalemate conditions are also detected, resulting in a tie message being displayed.

Special Moves

- **Castling**: The King and Rook may castle when no pieces occupy the squares between them and the King would not pass through check.
- **En Passant**: An Ant (Pawn) may capture an adjacent enemy Ant via en passant when the conditions are met.
- **Pawn Promotion**: When an Ant reaches the back rank of the board, the player is prompted to promote it to a Queen, Anteater, Bishop, Knight, or Rook.

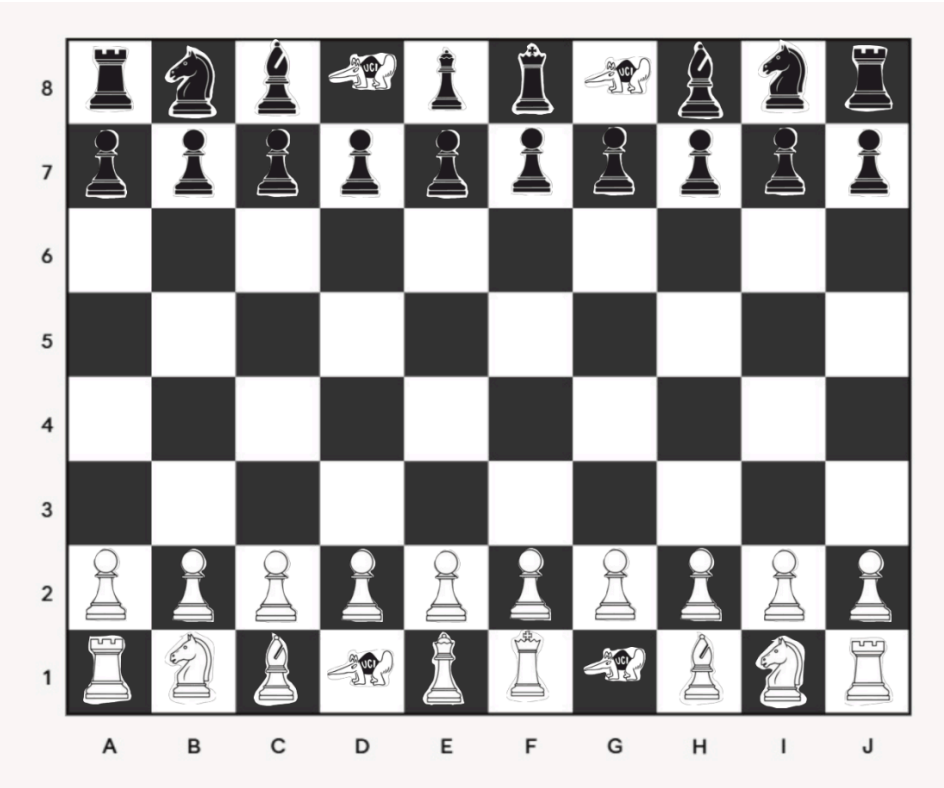
Undo Functionality

- Players may undo their last move, restoring any captured pieces to the board and adjusting point totals accordingly.
- Undo is only available on the player's own turn and handles all special move types, including castling and en passant.

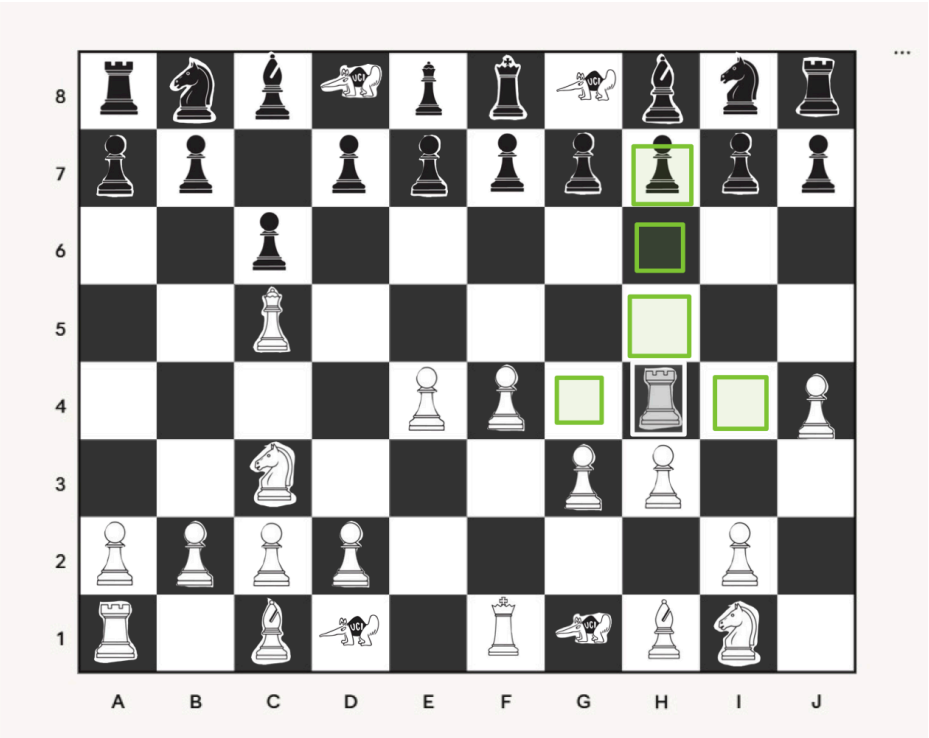
New Game and End-of-Game Summary

- At the conclusion of a game, an end message is displayed identifying the winner (or announcing a stalemate), along with a summary of moves played, hints used, and undos used.
- The player may start a new game or quit the application directly from the end-of-game screen.

Reference 1.31: Anteater Chess Piece



Reference 1.32: 8x10 Chessboard with Pieces



Reference 1.33: Highlighting Moves using Rook

2.1 System Requirements

Before installing and running the program please make sure your system has and meets the following requirements:

- Linux operating system
- GCC compiler
- Make Utility
- Ability to use the Linux terminal

2.2 Setup and Configuration

In order to install the program, please follow the steps below:

Step 1: Extract Files

a. To extract the files, type the following into your terminal:
"gtar xvzf Chess_V1.0_src.tar.gz"

Step 2: Change the Directory:

a. After you have finished extracting the files, change into the program directory. To change into the program directory, type the following into your terminal:
"cd chess"

Step 3: Compile the program:

a. Once you have changed to the correct directory, you can now compile the program. To compile the program, type the following into your terminal:
"make"

Step 4: Run the Program:

a. The last and final step is to run the program. To run the program, type the following into your terminal:
"./chess"

2.3 Uninstalling

In order to remove the program from your system, please follow the steps below:

Step 1: Clean the File:

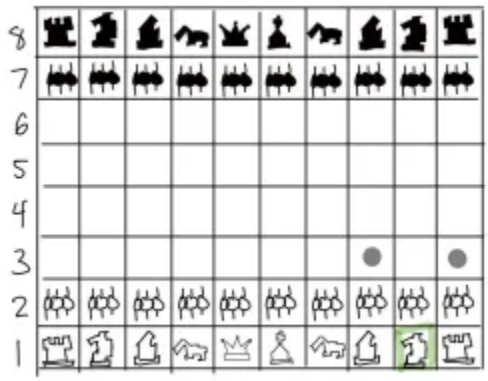
a. The first step is to clean the compiled files. In order to clean the compiled files, type the following into your terminal:
"make clean"

Step 2: Remove the Files:

a. Finally, to remove the files from your system, type the following into your terminal:
"rm-rf chess"

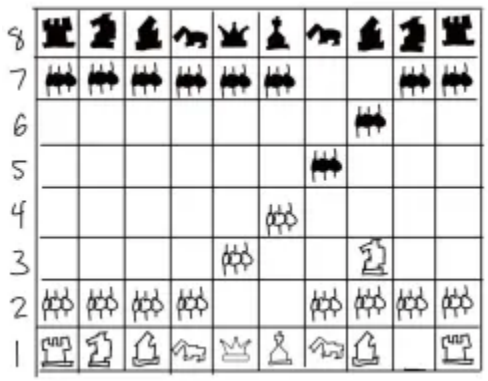
Function 1: Move

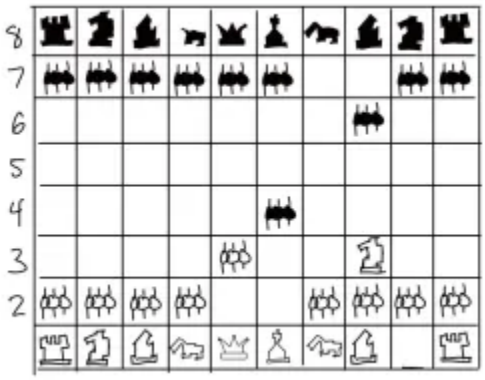
- Moves a piece from one spot to another depending on the piece type and the rule of the game
- **Inputs**: The user clicks on a piece and clicks where they want to move, piece color, turn
- **Outputs**: moves the chosen piece to the chosen location, calls the capture function if necessary, adds the move to the action text file
- **Logic**: checks which color the piece belongs to, verifies if the move is legal, checks for check
- **Error**: If move isn't legal, the piece will not move, (Text message will be prompted…TBD)



Function 2: Capture

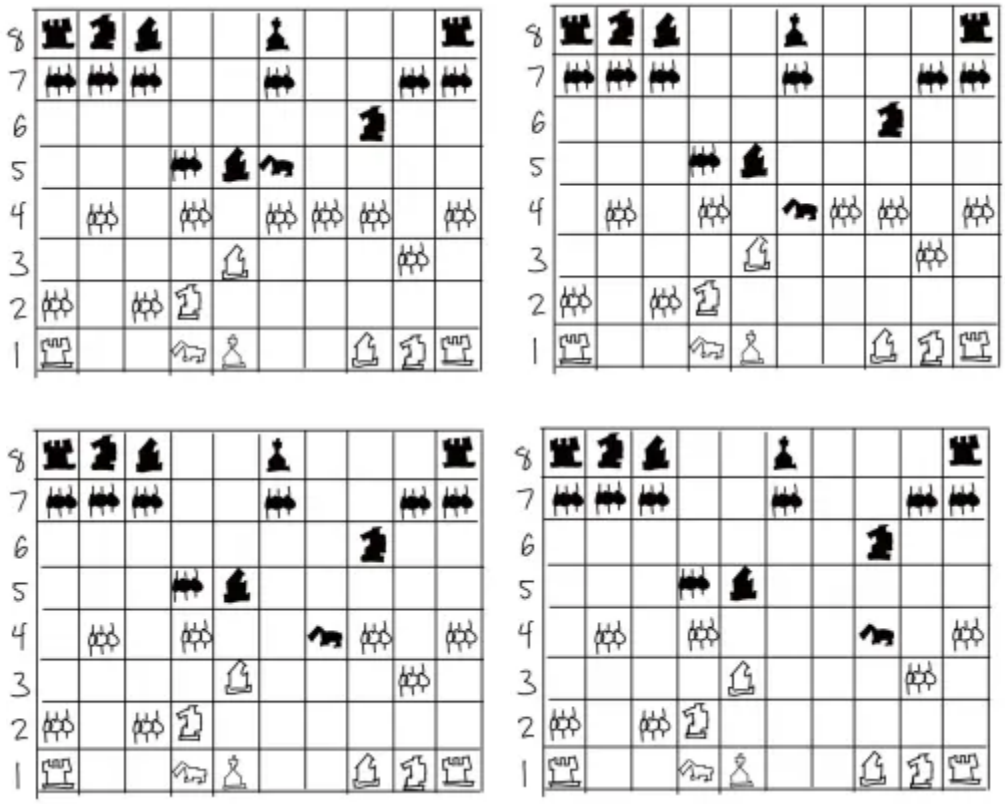
- Captures a piece and removes the captured piece from the board
- **Inputs**: takes move function and removes a captured piece
- **Outputs**: removes a captured piece, calls point counter, and adds name of the piece to a text file, adds the capture to the action text file
- **Logic**: removes the captured piece from the board simultaneously with the move function





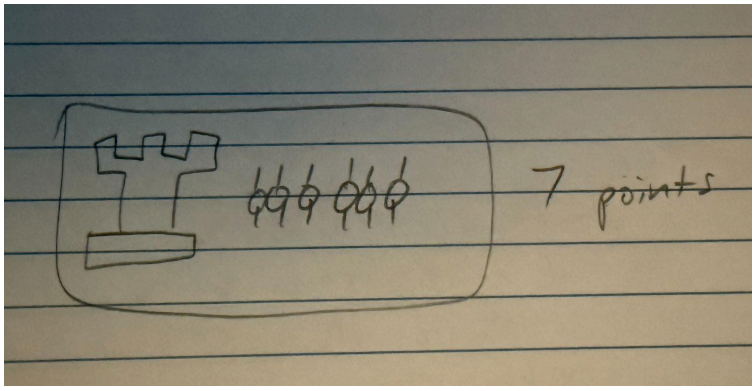
Function 3: Anteater Capture

- Captures pieces and removes the captured ant piece(s) from the board
- **Inputs**: takes move function of anteater, piece color
- **Outputs**: removes a captured ant piece, calls point counter, adds name of the piece to a text file, returns a boolean value if it can capture again, adds the capture to the action text file
- **Logic**: removes the captured piece from the board simultaneously with the move function, checks if another ant piece is orthogonally adjacent



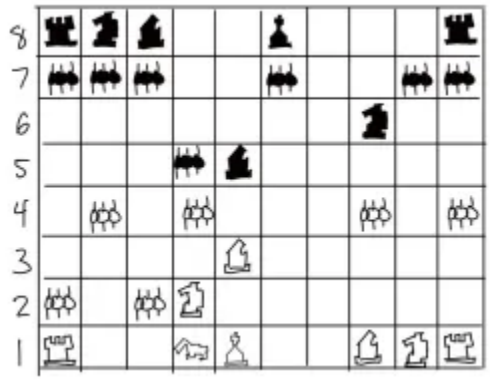
Function 4: Point Counter

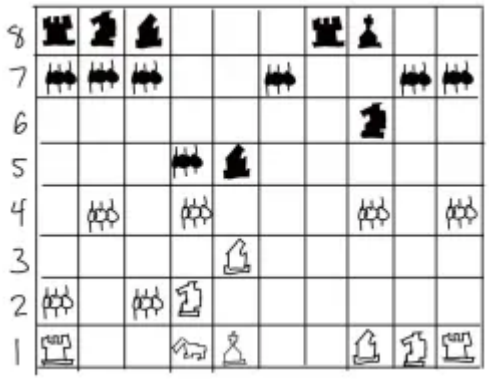
- Adds point depending on captured piece
- **Inputs**: captured piece point value, piece color
- **Outputs**: adds the points of captured piece to the total value in the points



Function 5: Castle

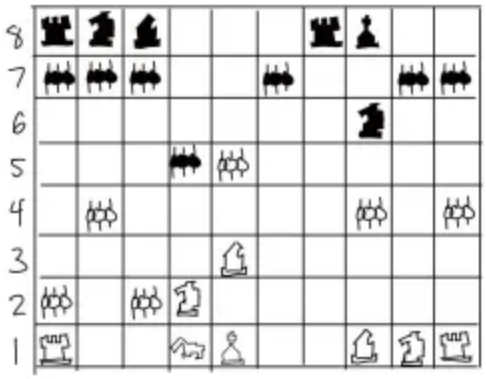
- Switches position of the king and a rook
- **Inputs**: move function, position of the king, position of the rook, piece color
- **Outputs**: moves the king two spaces closer to the rook and puts the rook one space on the other side of the king, adds the move to the action text file
- **Logic**: checks if the spaces between rook and king are empty, checks if the king will be in check if castle is complete, checks the direction the king is moving, finds the new location of the king and rook
- **Error**: If move isn't legal or the king will be in check, the piece will not move, (Text message will be prompted…TBD)

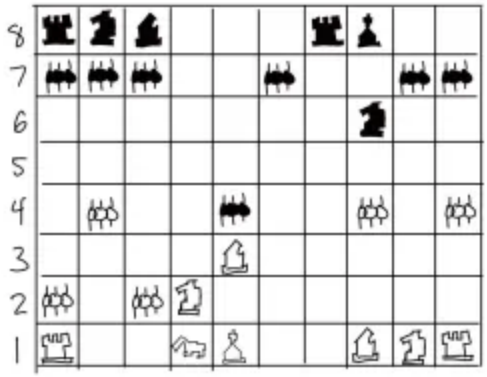




Function 6: En Passant

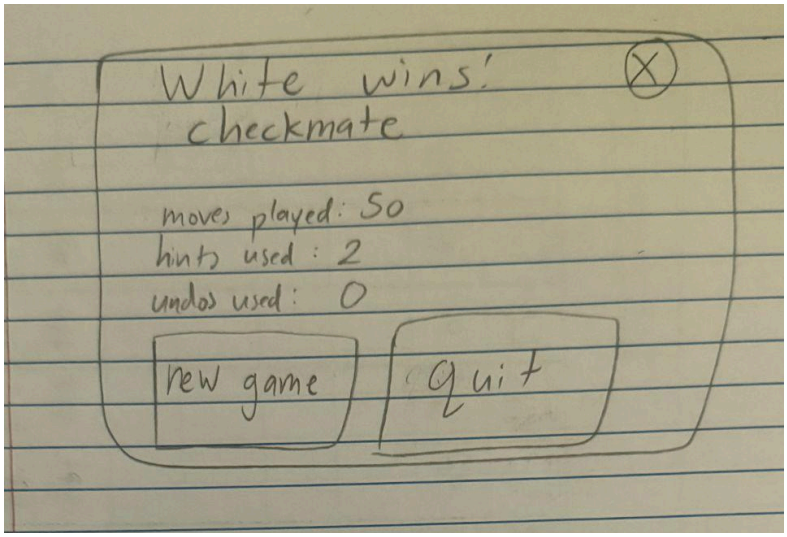
- Ant (pawn) piece captures an enemy ant (pawn) piece that is directly to the side the side of it
- **Inputs**: move function, position of ant, piece color
- **Outputs**: removes a captured piece, calls point counter, and adds name of the piece to a text file, adds the capture to the action text file
- **Logic**: check if a white ant is next to a black ant, check if the move space is open





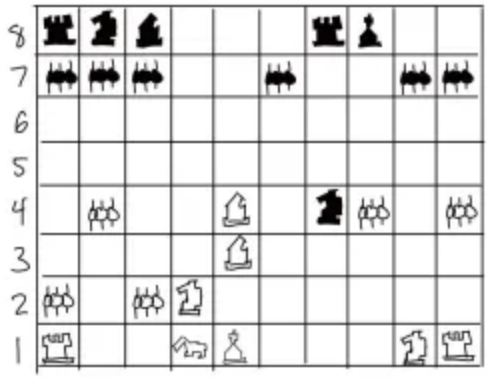
Function 7: End Message

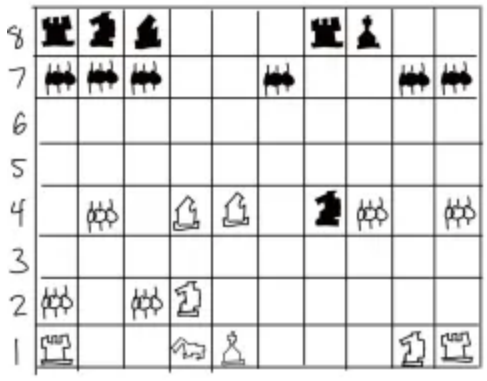
- Display who the winner is
- **Inputs**: checkmate/stalemate, piece color
- **Outputs**: display a message saying who the winner is
- **Logic**: checks for checkmate, see who is checkmated, checks for stalemate, displays winning message or tie message



Function 8: Check

- Checks for check, checkmate, or stalemate
- **Inputs**: move function, piece color, king position
- **Outputs**: displays which color is in check, returns boolean of true or false for checkmate or stalemate
- **Logic**: called after every move, sees which piece is attacking the king, if the king can't move to any spaces or no other piece can block, calls checkmate. If no move can be made by a player, stalemate will be called.





Function 9: Timer/change turn

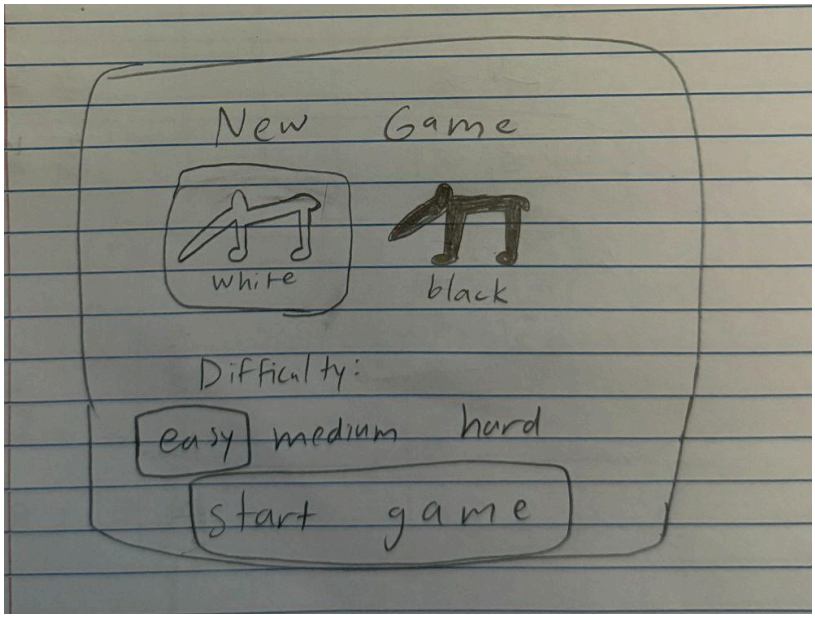
- Timer for player when it is their turn
- **Inputs**: New game function, turn, move function
- **Outputs**: timer counts down, changes turn, stops timer
- **Logic**: checks if the game has started, checks if who's turn it is, timer decrement until move function is called

Function 10: Turn

- Tells the game whose turn it is
- **Inputs**: turn, new game
- **Outputs**: determines who can do the next action
- **Logic**: checks to see if new game is called, checks to see whose turn it is

Function 11: New Game

- Starts a new game
- **Inputs**: user clicks the button to start a new game, difficulty, choose color
- **Outputs**: game starts, changes turn to white



Function 12: Undo

- Undoes the last move
- **Inputs**: turn, move, capture, ant capture, point, castle, en passant
- **Outputs**: Moves the moved piece back to its original position and reads the captured piece back to the board, removes the list of captured piece from text file, removes added points
- **Logic:** moves the moved piece back, checks to see if capture, ant capture, castle, or en passant was called, put the piece back, can only be used on who's turn it is
- **Error:** If undo is used by player if its not their turn, nothing will happen, (error message will be displayed…TBD)

Function 13: get position

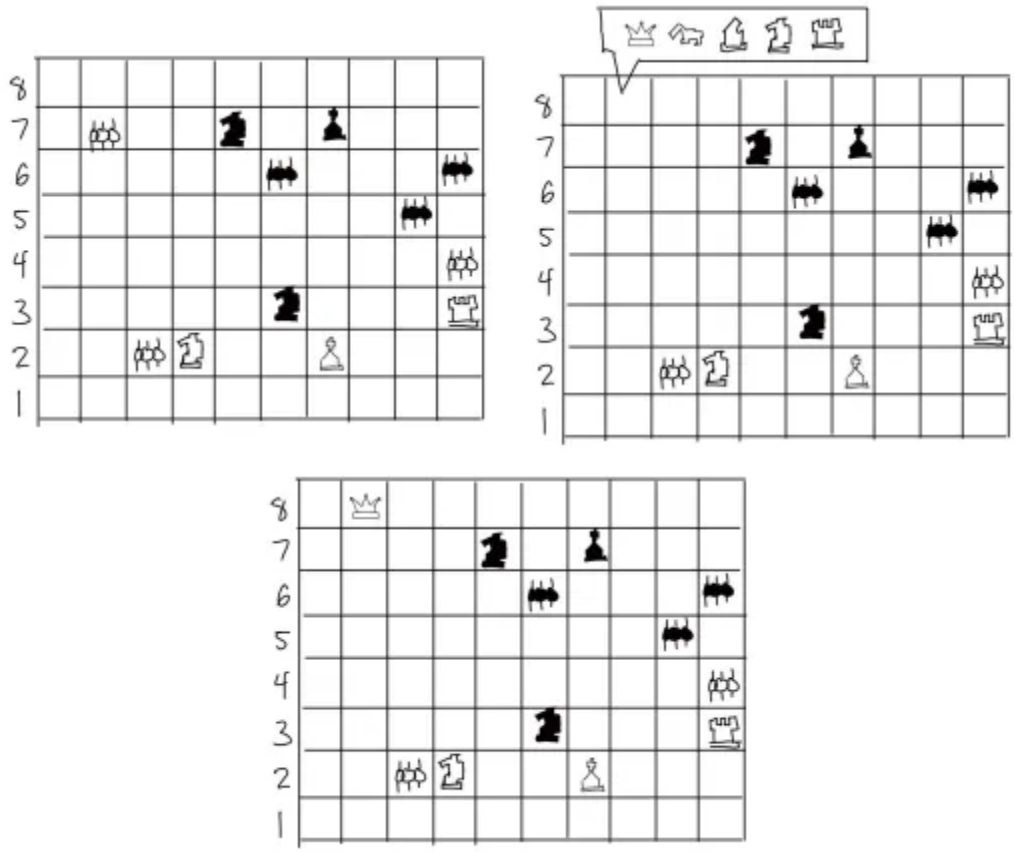
- Finds the position of a piece
- **Inputs**: piece, piece color
- **Outputs**: position of a piece

Function 14: Set position

- Changes the position of a piece
- **Inputs**: piece, piece color, new position
- **Outputs**: changes the position of

Function 15: Promotion

- Promote an ant (pawn) to a queen, anteater, bishop, knight, or rook
- **Inputs**: move, position, user input (user clicks on which piece they want)
- **Outputs**: calls promote function, calls check function



Function 16: promote

- Changes an ant (pawn) to a queen, anteater, bishop, knight, or rook
- **Inputs:** promotion function
- **Outputs:** pawn becomes another piece
- **Logic:** removes the current pawn and puts a piece of user choice in the location of the pawn

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ERROR MESSAGES

REFERENCE OF ERROR MESSAGES THAT MAY BE DISPLAYED DURING PROGRAM RUNTIME

ERROR MESSAGE	CAUSE	RESOLUTION

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